

The RANDOMRHS test set 2010 is a collection of 58 randomly generated instances of two-stage stochastic programs with pure integer variables in both stages and random right-hand sides. The problem formulation as well as the random instance generation scheme are discussed in Trapp et al. (submitted in 2012).

DATA:

The RANDOMRHS data contains a zip file for each of 26 test instance classes. For each IC-TX instance class, there are 3 instances in SMPS format, and for each IC-KX instance class, there is a single instance in SMPS format. The following table presents the characteristics of these 26 instance classes. Denote $|\Omega|$ as the number of scenarios, m as the number of (combined) first- and second-stage constraints, and n_1, n_2 as the number of first-stage and second-stage integer variables, respectively. Δ refers to the density of technology matrix T and recourse matrix W , “Rows” and “Cols” refer to the number of rows and columns in the extensive form. $|\mathbf{B}^1|$ and $|\mathbf{B}^2|$ refer to the search space size, which is itself a function of m and right-hand side magnitudes.

	$ \Omega $	m	n_1	n_2	Δ	Extensive Form		Search Space Size	
						Rows	Cols	$ \mathbf{B}^1 $	$ \mathbf{B}^2 $
IC-T1	1,000	20	50	50	0.3	2.0E4	5.0E4	3.7E15	6.7E20
IC-T2	1,000	20	50	50	0.4	2.0E4	5.0E4	3.7E15	6.7E20
IC-T3	1,000	20	50	50	0.5	2.0E4	5.0E4	3.7E15	6.7E20
IC-T4	5,000	50	100	100	0.3	2.5E5	5.0E5	8.1E38	1.2E52
IC-T5	5,000	50	100	100	0.4	2.5E5	5.0E5	8.1E38	1.2E52
IC-T6	5,000	50	100	100	0.5	2.5E5	5.0E5	8.1E38	1.2E52
IC-T7	8,000	100	200	200	0.3	8.0E5	1.6E6	6.5E77	1.4E104
IC-T8	8,000	100	200	200	0.4	8.0E5	1.6E6	6.5E77	1.4E104
IC-T9	8,000	100	200	200	0.5	8.0E5	1.6E6	6.5E77	1.4E104
IC-T10	10,000	100	300	300	0.3	1.0E6	3.0E6	6.5E77	1.4E104
IC-T11	10,000	100	300	300	0.4	1.0E6	3.0E6	6.5E77	1.4E104
IC-T12	10,000	100	300	300	0.5	1.0E6	3.0E6	6.5E77	1.4E104
IC-T13	10,000	20	50	50	0.5	2.0E5	5.0E5	3.7E15	6.7E20
IC-T14	50,000	50	100	100	0.4	2.5E6	5.0E6	8.1E38	1.2E52
IC-T15	80,000	100	200	200	0.3	8.0E6	1.6E7	6.5E77	1.4E104
IC-T16	300,000	250	1000	500	0.3	7.5E7	1.5E8	3.4E194	2.2E260
IC-K1	7,812	6	1000	100	0.5	4.7E4	7.8E5	4.7E4	1.0E6
IC-K2	7,812	6	500	500	0.5	4.7E4	3.9E6	4.7E4	1.0E6
IC-K3	7,812	6	300	500	0.5	4.7E4	3.9E6	4.7E4	1.0E6
IC-K4	7,812	6	500	500	0.5	4.7E4	3.9E6	4.7E4	1.0E6
IC-K5	7,812	6	500	500	0.5	4.7E4	3.9E6	4.7E4	1.0E6
IC-K6	7,812	6	500	500	0.8	4.7E4	3.9E6	4.7E4	1.0E6
IC-K7	7,812	7	1000	500	0.7	5.5E4	3.9E6	2.8E5	1.9E7
IC-K8	7,812	7	500	500	0.7	5.5E4	3.9E6	2.8E5	1.9E7
IC-K9	7,812	7	300	500	0.7	5.5E4	3.9E6	2.8E5	1.9E7
IC-K10	7,812	7	1000	500	0.3	5.5E4	3.9E6	2.8E5	1.9E7

SOLUTION:

The following table presents the overall optimal objective values of 56 of the 58 instances, where two of the test instances (IC-K2 and IC-K10) were not solved within the threshold of 200 hours. It also presents the CPU times using the global branch-and-bound algorithm combined with the *First-Stage Exterior Bounding* approach (Variant E1), as presented in Trapp et al. (2012). The instances were compiled using Microsoft Visual Studio 2005 on a PC equipped with Windows XP, a dual-core Intel 3GHz processor, 3GB of RAM, and IBM ILOG CPLEX 11.0.

Test Instance	Obj	CPU Time (s.)	Test Instance	Obj	CPU Time (s.)
IC-T1-1	42.5979	119118.0	IC-T10-2	12.5578	79641.1
IC-T1-2	58.8979	154371.0	IC-T10-3	11.4349	51218.4
IC-T1-3	36.5027	18768.7	IC-T11-1	11.0000	4517.2
IC-T2-1	22.5313	821.5	IC-T11-2	10.0005	2989.3
IC-T2-2	21.9742	723.2	IC-T11-3	10.0000	4084.2
IC-T2-3	22.7124	1102.7	IC-T12-1	6.0000	2335.4
IC-T3-1	12.9856	68.1	IC-T12-2	6.0129	2426.1
IC-T3-2	13.4385	83.7	IC-T12-3	7.0000	2312.6
IC-T3-3	14.9139	113.6	IC-T13-1	15.9902	1227.5
IC-T4-1	24.5262	19901.2	IC-T13-2	16.3746	765.3
IC-T4-2	20.5196	23004.7	IC-T13-3	15.5797	1108.1
IC-T4-3	25.0739	58118.7	IC-T14-1	14.0000	7641.0
IC-T5-1	14.0000	1147.9	IC-T14-2	11.4615	7525.2
IC-T5-2	14.0689	1088.9	IC-T14-3	17.0047	12280.3
IC-T5-3	12.0080	662.8	IC-T15-1	16.0011	96552.6
IC-T6-1	10.4273	196.3	IC-T15-2	14.2949	93082.6
IC-T6-2	10.5991	283.8	IC-T15-3	14.0049	103870.0
IC-T6-3	10.3598	279.0	IC-T16-1	7.4796	549179.0
IC-T7-1	11.4057	8225.8	IC-T16-2	7.0000	533165.0
IC-T7-2	12.0000	8674.5	IC-T16-3	7.0308	555403.0
IC-T7-3	11.3325	7007.7	IC-K1	86.9666	13563.8
IC-T8-1	6.9461	1389.2	IC-K3	81.1436	43031.8
IC-T8-2	7.0039	1333.5	IC-K4	195.7830	24280.5
IC-T8-3	8.0000	1136.4	IC-K5	93.5154	188537.0
IC-T9-1	5.0001	851.2	IC-K6	26.3009	13993.1
IC-T9-2	5.0001	874.4	IC-K7	42.4066	174291.0
IC-T9-3	6.0001	839.5	IC-K8	49.7239	208254.0
IC-T10-1	12.7890	107086.0	IC-K9	29.3508	44349.8

REFERENCE:

Andrew C. Trapp, Oleg A. Prokopyev, Andrew J. Schaefer, 2012. "On a Level-Set Characterization of the Integer Programming Value Function and its Application to Stochastic Programming," Submitted to *Operations Research*.